

Auteur: X Tenka Tenzin Choezin

Studierichting: Informatica

Oefeningenreeks 1D nr. 20.2

$$z^4 + z^2 - 12$$

$$\text{Stel } t = z^2 \Rightarrow t^2 + t - 12$$

Nulpunten berekening

$$D = 1 - 4 \cdot 1 \cdot (-12) = 49 \Rightarrow \sqrt{49} = 7$$

$$t_1 = \frac{-1+7}{2} = 3$$

$$t_2 = \frac{-1-7}{2} = -4$$

$$\text{Door } t = z^2 \Leftrightarrow z = \sqrt{t}$$

$$z_{1,2} = \pm\sqrt{3}$$

$$z_{3,4} = \pm 2i$$

Ontbinden van de veelterm

$$(z^2 + 4)(z + \sqrt{3})(z - \sqrt{3})$$

References